



先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 06 – Feature Engineering: Transformation, Construction, and Selection



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A hospital table can lie without changing a value

Raw row

```
age=67, ward=B, test_time=14:30, discharge_code=home, lab_value
missing
```

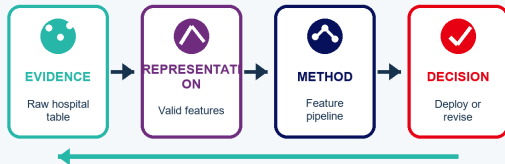
Hidden problem

The discharge code is recorded after the prediction decision.

Question

Which columns are available at the exact prediction time?

THIS LECTURE'S DECISION LOOP



Outcome and error feed the next observation

Which representation makes structure learnable without leakage?

Syllabus allocation for Lecture 06

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Syllabus section	Time	What students must do
Feature transformation	1.0 hour	fit and apply valid representations
Feature construction	0.5 hour	turn domain hypotheses into variables
Feature selection	1.5 hours	compare filter, wrapper, and embedded methods

Course boundary

Principal component analysis belongs to the later dimensionality reduction lecture.

Feature engineering sits between raw data and learning

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FIT ONLY ON TRAINING EVIDENCE



The same fitted rules transform future rows



Raw data. Measurements, categories, text, time stamps, and missingness.

Feature pipeline. Transform, construct, and select using training data only.

Learner. Fit a model and evaluate it on untouched evidence.

Feature type determines valid operations

Raw type	Examples	Valid first questions
Continuous	temperature, price	scale, skew, outliers
Ordinal	pain level, grade	order and spacing
Nominal	ward, color	encoding and unknown values
Time	date, hour	cycle and prediction cutoff

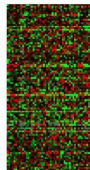


face images

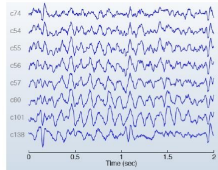
Zambian President Levy Mwanawasa has won a second term in office in an election his challenger Michael Sata accused him of rigging, official results showed on Monday.

According to media reports, a pair of hackers said on Saturday that the Firefox Web browser, commonly perceived as the safer and more customizable alternative to market leader Internet Explorer, is critically flawed. A presentation on the flaw was shown during the ToorCon hacker conference in San Diego.

documents



gene expression data



MEG readings

Fit and transform are different actions

Fit

Learn parameters such as means, medians, category sets, or bin edges from training data.

Transform

Apply the learned rule without recalculating it on validation or test data.

Leakage

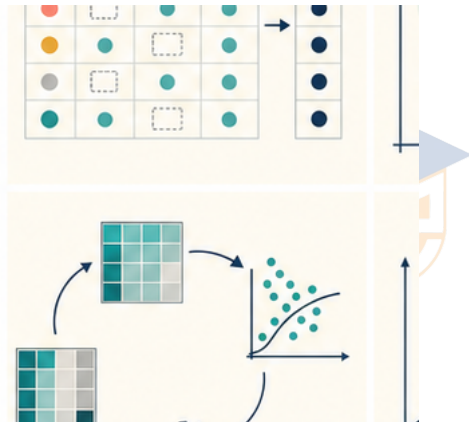
Calling fit on the test set lets future evidence influence the representation.

The data split comes before learned preprocessing

- 1 Reserve the test set.
- 2 Fit preprocessing on the training portion.
- 3 Apply the same fitted preprocessing to validation and test rows.
- 4 Fit and select the model without reopening the test set.

Order matters

Split first, then learn medians, scales, categories, and selected features.



t information
 observed context
 putation can bias

Reasonable when missingness is rare and approximately random.

Reasonable when a field is mostly unavailable or operationally unreliable.

Deleting rows can remove a specific patient group rather than random noise.

Imputer	Suitable situation	Failure
Mean	roughly symmetric numeric data	moves with outliers
Median	skewed numeric data	ignores subgroup structure
Mode	categorical or discrete data	can exaggerate the majority

Imputation learns training statistics

$$\tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \text{ observed} \\ m_j, & x_{ij} \text{ missing} \end{cases}$$

Training estimate

m_i is learned from the training fold only.

Test time

Reuse the same value even when the new batch has a different median.

A missing indicator can preserve process information

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Construct

$m_j = 1$ when feature j is missing and 0 otherwise.

Clinical process

A test was not ordered because the initial examination looked normal.

Caution

The reason for missingness may change when hospital policy changes.

A histogram comes before a transformation

Inspect

Look for skew, multiple modes, truncation, and impossible values.

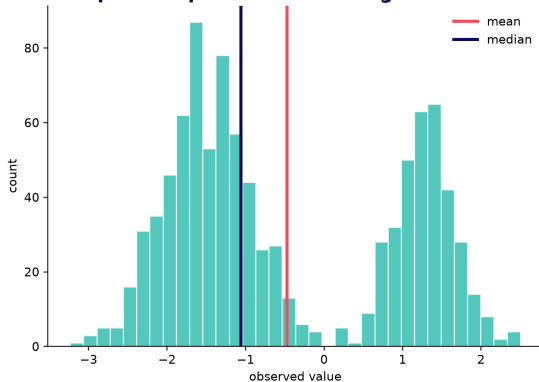
Question

Does one peak represent one population or two mixed subgroups?

Do not beautify blindly

A smoother distribution is not automatically more predictive.

Inspect shape before choosing a transform



Outliers may be errors or rare truths

Error

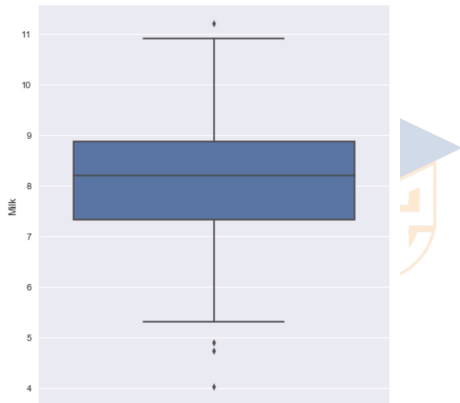
Body temperature recorded as 370 instead of 37.0.

Rare truth

An unusually high transaction that is a genuine fraud signal.

Decision

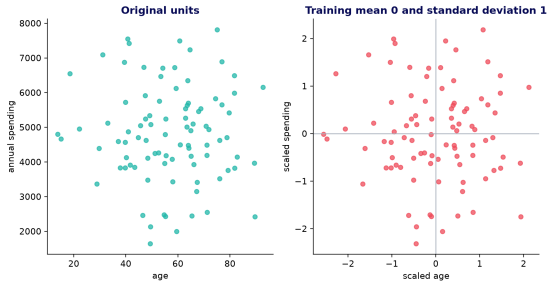
Validate the record and its units before clipping or removing it.



$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

Estimate μ_j and σ_j from the training fold.

One unit of z means one training standard deviation.



Min max scaling fixes a training range

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$$x'_j = \frac{x_j - \min_{\text{train}} x_j}{\max_{\text{train}} x_j - \min_{\text{train}} x_j}$$

Training values

Map the observed training minimum to 0 and maximum to 1.

Future value

A new value outside the training range can map below 0 or above 1.

Robust scaling uses quantiles

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$$x'_j = \frac{x_j - \text{median}(x_j)}{Q_{0.75}(x_j) - Q_{0.25}(x_j)}$$

Benefit

Median and interquartile range react less to extreme observations.

Tradeoff

Quantiles still come from training data and can shift across populations.

Row normalization answers a different question

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$$x'_i = \frac{x_i}{\|x_i\|_2}$$

Across columns

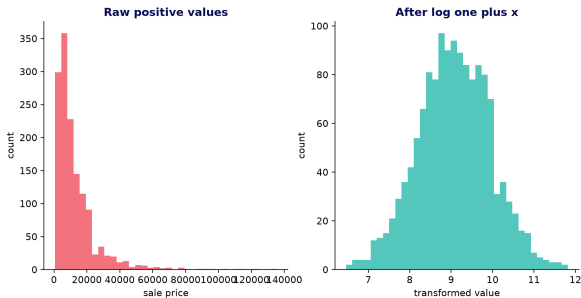
Standardization rescales each feature using many samples.

Across one row

Normalization rescales one sample vector to unit length.

Log transforms compress right tails

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$$x' = \log(1 + x)$$

Effect. Large values are compressed more than small values.

Use case. Price, counts, and response times often span several orders of magnitude.

Power transforms include log as one case

Family

Box Cox for positive values and Yeo Johnson for values that may include zero or negatives.

Learned parameter

The power parameter must be fitted inside the training fold.

Interpretation

Automatic Gaussian looking output does not guarantee better prediction.

Quantile bins must be learned on training data

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Equal frequency idea

Choose bin edges so training bins contain similar counts.

Leakage

Computing quantiles on all rows lets validation and test distributions set the boundaries.

Deployment

Future values use the stored training edges even when bin counts become unequal.

Ordinal and nominal categories need different encodings

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Type	Example	Possible encoding
Ordinal	low, medium, high	ordered integer with verified order
Nominal	red, green, blue	one hot or another order free encoding



Distance

An integer code for nominal values invents an order and spacing.

One category column may be redundant because all indicators sum to one.

Keeping all columns may be acceptable and easier to interpret.

The dropped category becomes the baseline for coefficient comparisons.

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Ward codes A, B, and C are observed.

A new ward D appears after reorganization.

Ignore into an all zero code, map to other, or retrain under a controlled update.

High cardinality makes one hot sparse

Examples

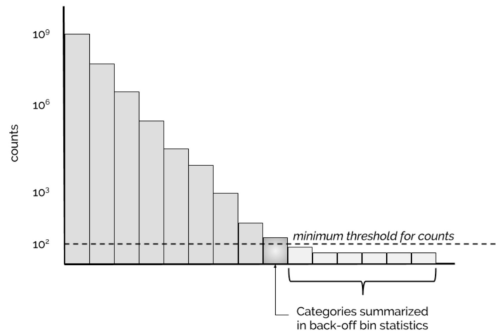
User identifier, product identifier, postal code, and device identifier.

Statistical problem

Many categories have too few examples to estimate reliable effects.

Computational problem

The design matrix becomes very wide and sparse.



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No target label is used and one column can replace many indicators.

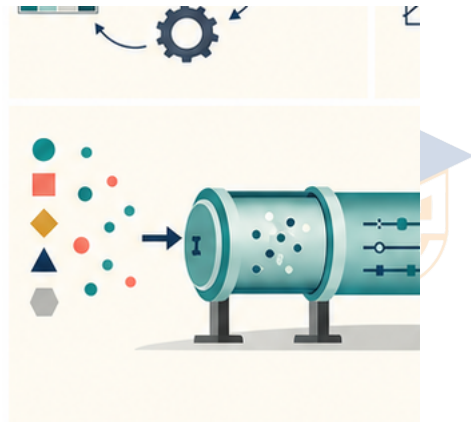
Different categories with the same frequency receive the same value.

Out of fold target encoding separates roles

- 1 Split the training data into folds.
- 2 For one fold, compute category statistics from the other folds.
- 3 Encode the held out fold without its own labels.
- 4 Repeat and then fit the predictive model.

Outer validation

The whole encoder still belongs inside the outer cross validation loop.



- Calendar: year, month, weekday, and holiday.
- Clock: hour and minute.
- Elapsed time: age of an event or time since the previous event.
- Cutoff: information available before prediction.

Raw timestamp

A large epoch number rarely expresses the right relationships by itself.

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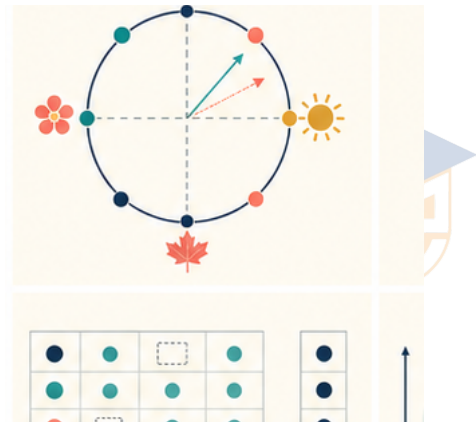
$$h_{\cos} = \cos(2\pi h/24)$$

Geometry

Hour 23 and hour 0 become nearby points on a circle.

Period

Use the correct cycle length for the domain.



Counts, term weighting, or learned embeddings convert tokens into numeric representations.

Pixels, engineered descriptors, or learned representations become feature vectors.

Detailed representation learning appears in the deep learning lectures.



```
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler

numeric = make_pipeline(SimpleImputer(strategy="median"),
                        StandardScaler())
categorical = make_pipeline(SimpleImputer(strategy="most_frequent"),
                           OneHotEncoder(handle_unknown="ignore"))
prep = ColumnTransformer([
    ("num", numeric, numeric_columns),
    ("cat", categorical, categorical_columns),
])
```

Contract

Calling fit on the full pipeline learns every preprocessing parameter from training data.

Measurements are stored for operations, not necessarily for prediction.

A relationship may matter more than either field alone.

Express that relationship with a reproducible formula.



A taxi estimate starts with a map question

Raw inputs

Pickup latitude, pickup longitude, dropoff latitude, dropoff longitude, and departure time.

Useful relation

Travel distance and traffic period are closer to the mechanism of trip duration.

Unavailable value

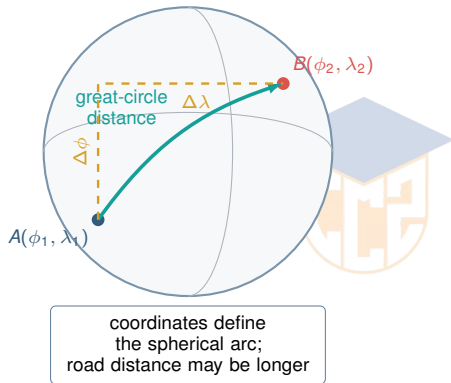
Actual arrival time cannot be used when the estimate is made.

Interpretation

Four geographic coordinates become one approximate great circle distance.

Model limits

Road distance can differ because of rivers, bridges, and street layout.



Ratios reveal scale free behavior

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$$\text{occupancy} = \frac{\text{used capacity}}{\text{total capacity}}$$

Hospital beds

Forty occupied beds mean different pressure in a 50 bed and a 500 bed hospital.

Denominator

Near zero denominators create unstable or extreme ratios.

A temperature rise from a personal baseline can matter even when the absolute value is moderate.

100

A baseline measured after the event would leak future information.

$$(x_1, x_2) \mapsto (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$$

A linear learner in the expanded space can represent curved relations.

Degree and dimension make the number of terms rise quickly.

Feature count grows before information grows

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$$p_{\leq d} = \binom{p+d}{d} - 1$$

Ten inputs

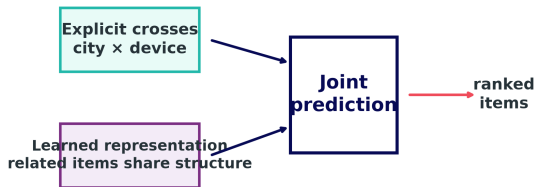
Degree three expansion produces 285 nonconstant terms.

Sample size

More columns do not create more independent observations.

Wide and deep models use two kinds of memory

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Wide part. Memorizes explicit crosses such as city and device combinations.

Deep part. Learns distributed representations that can generalize to related combinations.

Recommendation. Known purchase patterns and unseen but similar items need different strengths.

Elapsed time is often more useful than a date

$$\text{tenure} = t_{\text{prediction}} - t_{\text{start}}$$

Subscription

Account age carries a direct behavioral meaning that a signup date does not.

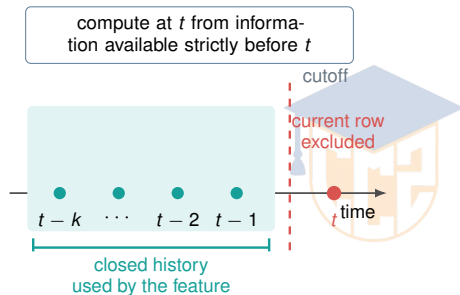
Negative values

A negative duration usually signals a timestamp error or leakage.

$$m_i(t) = \frac{1}{k} \sum_{s=t-k}^{t-1} x_i(s)$$

Count transactions in the seven days before the current payment.

Including the event being predicted can reveal part of the answer.



Group aggregates can identify context or identity

Context

A store's historical average demand can describe its operating environment.

Identity shortcut

A nearly unique group statistic may let the model memorize an entity.

Cold start

A new store has no personal history.

Future leakage survives elegant code

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```
orders = orders.sort_values("time")
orders["past_mean"] = (
    orders.groupby("customer")["amount"]
        .transform(lambda s: s.shift(1).rolling(10).mean())
)
```

Critical operation

shift(1) excludes the current order before the rolling mean is computed.

Feature hashing trades memory for collisions

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$$j = h(c) \bmod m$$

Benefit

Map a huge or open category set into a fixed number of columns.

Collision

Different categories can share a column and their effects become entangled.

Missingness can interact with context

$$z = \mathbf{1}(x \text{ missing}) \times \mathbf{1}(\text{night shift})$$

Operations

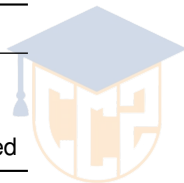
A missing lab result at night may reflect staffing rather than patient physiology.

Policy drift

The interaction can change when staffing or ordering rules change.

Units belong in the feature definition

Feature	Bad definition	Operational definition
speed	distance divided by time	kilometres divided by hours
dosage	dose divided by weight	milligrams divided by kilograms
price change	new minus old	currency and time window stated



A custom transformer makes a hypothesis reproducible

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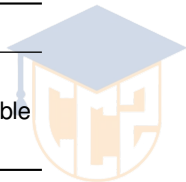
```
from sklearn.base import BaseEstimator, TransformerMixin

class RatioFeature(BaseEstimator, TransformerMixin):
    def fit(self, X, y=None):
        return self
    def transform(self, X):
        out = X.copy()
        out["occupancy"] = out["used"] / out["capacity"].clip(lower=1)
        return out
```

Benefit

The same formula becomes part of fitting, cross validation, and deployment.

Model input	Validation score	Decision
raw fields	baseline	keep as reference
raw plus proposed feature	compare across folds	keep only if gain is stable
proposed feature alone	diagnostic	check for shortcuts



For a delivery delay model, propose one ratio, one time feature, and one interaction.

load ratio, time since dispatch, and rain multiplied by rush hour.

Each feature must be available before the promised delivery time.

1

Transform data

2

Construct features

3

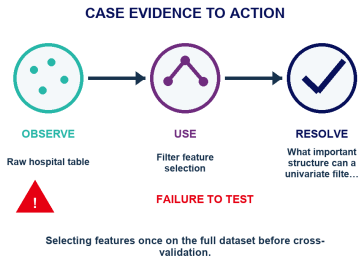
Compare selection methods

4

Protect pipelines

If it cannot be demonstrated, it is not yet a learning outcome.

Feature selection keeps a subset of existing columns



$$S^* = \arg \min_{S \subseteq \{1, \dots, p\}} \hat{R}(S) + \lambda \Omega(S)$$

Goal. Balance predictive evidence, complexity, stability, and operating cost.

Not extraction. Selection preserves columns. Extraction creates new coordinates.

Why remove features

Advanced features 数字工程研究中心
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- Reduce variance when samples are limited.
- Lower measurement and deployment cost.
- Shorten training and inference time.
- Remove redundant or unstable signals.
- Improve human review when selected variables retain meaning.

Myth

A smaller set is not automatically more accurate.

Selection and extraction preserve different things

Operation	Output	Meaning
feature selection	subset of original columns	original semantics remain
feature extraction	new combinations or coordinates	semantics can become mixed
feature construction	new domain variables	semantics are designed explicitly

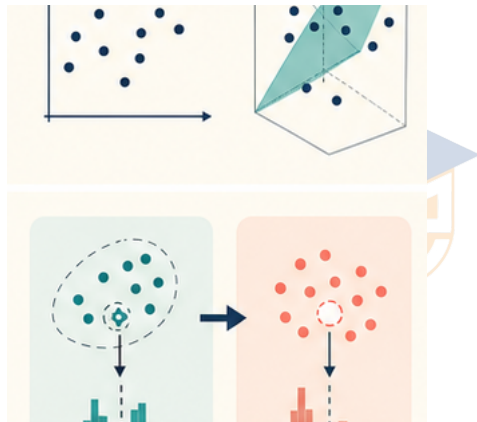
Later lecture

Principal component analysis is treated under dimensionality reduction.

Family	Question	Typical cost
filter	does a feature associate with the target or duplicate another feature	low
wrapper	does a subset improve a chosen model under validation	high
embedded	which features survive while the model is fitted	medium

- ## Leakage

Selecting once on the full data lets every validation label influence the candidate set.



Variance threshold removes constants

$$\widehat{\text{Var}}(x_j) < \tau \quad \Rightarrow \quad \text{remove } x_j$$

Use

Discard constant or nearly constant columns without using the target.

Scaling

The numerical threshold depends on feature units and encoding.

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$$

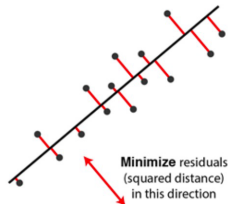
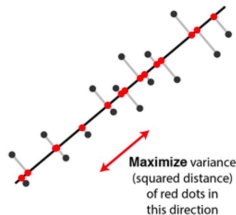
$-1 \leq r \leq 1$ with sign indicating direction.

Zero correlation does not imply independence.

Clusters of features carrying nearly the same linear signal.

Keep the cheaper, more stable, or more interpretable representative.

Redundancy among inputs is different from association with the target.



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$$I(X; Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

Estimation

Finite samples, continuous variables, and tuning choices affect the estimate.

One unusual category appears three times and all three labels are positive.

The apparent information may disappear in another sample.

Inspect support counts and selection frequency across folds.

Chi square tests categorical association

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$$\chi^2 = \sum_{r,c} \frac{(O_{rc} - E_{rc})^2}{E_{rc}}$$

Use

Compare observed and expected counts in a contingency table.

Requirements

Inputs should be nonnegative and expected cell counts should not be too small.

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$$F = \frac{\text{between group variation}}{\text{within group variation}}$$

Does a numeric laboratory value have different means across classes

Outliers, unequal variances, and strong nonnormality can distort the summary.

The exclusive OR problem defeats univariate filters

x1	x2	class
0	0	0
0	1	1
1	0	1
1	1	0

Observation

Neither feature is informative alone, yet the pair determines the class.

Text filters can reduce a huge vocabulary

Input

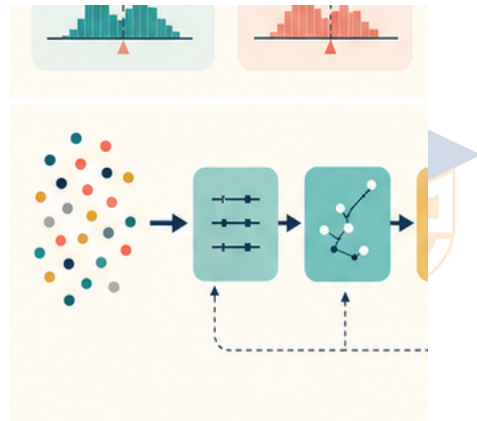
A document term matrix may contain tens of thousands of sparse columns.

Screen

Rank terms with chi square or mutual information using training labels.

Meaning

Selected words can reflect topic, source, or annotation artifacts.



100

100

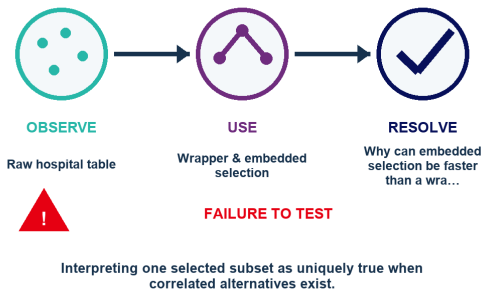
Interpretation

Selection frequency measures how often a feature survives resampling.

Correlation

Equivalent features may alternate, giving each a modest frequency.

CASE EVIDENCE TO ACTION



Domain review can veto a high score

Question	Example failure	Action
available at prediction	discharge code	remove
stable definition	hospital policy flag	monitor or remove
acceptable use	protected attribute proxy	fairness review
reasonable cost	expensive test	compare benefit with burden



Forward selection grows a model

- 1 Start with no selected features.
- 2 Try adding each remaining candidate.
- 3 Keep the addition with the best validated improvement.
- 4 Stop when improvement is too small or the budget is reached.

Greedy search

An early choice can block a better later combination.

Backward elimination starts with everything

- 1 Fit the model with all candidate features.
- 2 Try removing one feature at a time.
- 3 Drop the feature whose removal hurts validation least.
- 4 Continue until the stopping rule is met.

Feasibility

Starting with all columns may be impossible when dimension is huge.

The feature elimination follows



Ranking changes after every removal because the model is refitted on the remaining columns.

- 1 Fit a model on the current feature set.
- 2 Rank current features by a model specific importance.
- 3 Remove the weakest feature or group.
- 4 Refit and repeat until the target size is reached.

Model dependence. A linear weight and a tree importance express different preferences.

Wrapper methods spend models as currency

$$\text{cost} \approx \text{candidate subsets} \times \text{folds} \times \text{model fit time}$$

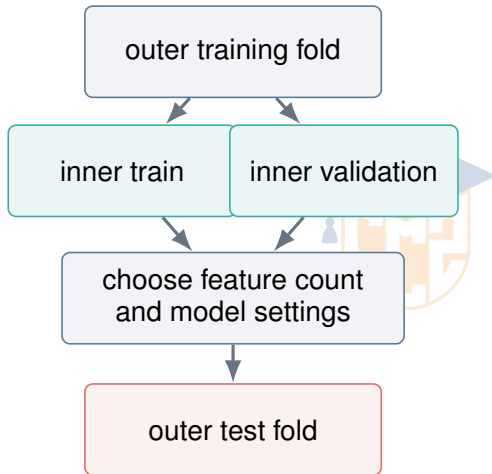
Scale

Testing 100 additions over 5 folds already requires hundreds of fits.

Budget

Use screening, step size, or early stopping when the search is large.

Open the test set only after the procedure has been fixed.



One norm regularization performs embedded selection

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$$\min_w \hat{R}(w) + \lambda \sum_{j=1}^p |w_j|$$

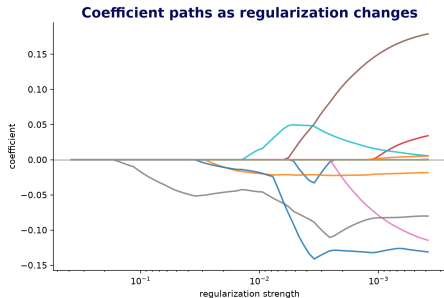
Effect

Some fitted coefficients become exactly zero.

Scale

Standardize comparable numeric inputs before comparing coefficient shrinkage.

Lasso paths show entry order and instability



Read the path. Each curve traces one coefficient as regularization changes.

Strong penalty. Most curves remain at zero.

Interpretation. Entry order can change when correlated features or samples change.

Trees select features through splits

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Embedded behavior

A feature not used in any split has zero split based importance.

Interaction

A feature can become useful only after an earlier branch conditions the data.

Instability

Correlated predictors may substitute for one another across trees.

Impurity importance can favor many cut points

$$I_j = \sum_{t: v(t)=j} \frac{n_t}{n} \Delta i_t$$

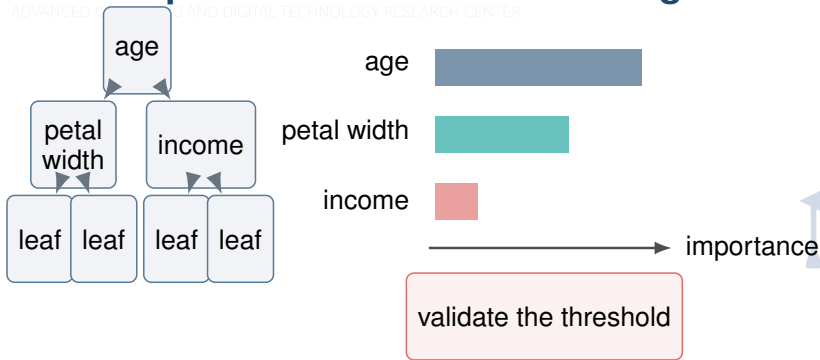
Meaning

Add weighted impurity decreases from all splits using feature j .

Bias

Continuous or high cardinality features offer more candidate cut points.

Tree selection keeps variables with validated gain



- 1 Fit the tree model inside a training fold.
- 2 Estimate feature importance without opening the validation labels.
- 3 Apply a threshold or select a count.
- 4 Evaluate the entire pipeline on the held out fold.

Threshold search. The importance threshold is another hyperparameter.

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Repeated permutations can be computationally expensive.

One program can compare selection families

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```
from sklearn.datasets import load_breast_cancer
from sklearn.feature_selection import SelectKBest, mutual_info_classif
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

X, y = load_breast_cancer(return_X_y=True)
pipe = make_pipeline(SelectKBest(mutual_info_classif, k=10),
                    StandardScaler(),
                    LogisticRegression(max_iter=3000))
print(cross_val_score(pipe, X, y, cv=5).mean())
```

Boundary

Selection and scaling are refitted inside every fold.

Candidate	Predictive evidence	Operational concern
routine vital sign	moderate and stable	cheap and always available
special laboratory test	slightly stronger	slow and costly
post treatment code	extremely strong	unavailable at prediction



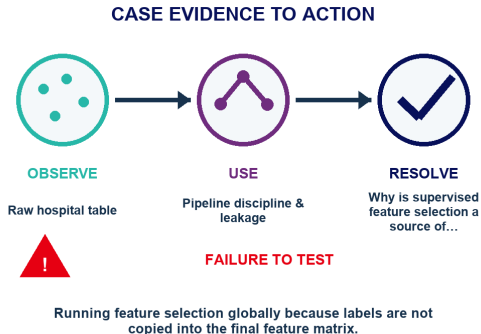
Remove the leaked code and compare the first two under a cost aware objective.

The full feature pipeline has one direction

- 1 Define prediction time and label.
- 2 Split data with the deployment structure.
- 3 Fit transformations and historical construction on training data.
- 4 Select features inside validation and fit the learner.
- 5 Evaluate once on untouched evidence.

No return path

Information from validation, test, or the future must not flow backward.



One raw row becomes a model vector

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Stage	Representation
raw	age 67, city Nanjing, time 23:00, income missing
transform	scaled age, one hot city, cyclic hour, imputed income, missing flag
construct	account age and recent activity ratio
select	retain validated and available columns

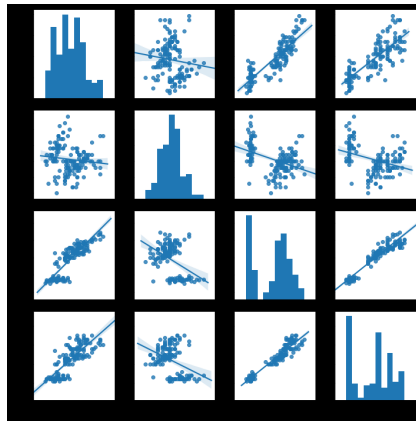


Bug hunt in a polished notebook

```
df["income"] = df["income"].fillna(df["income"].median())
X = pd.get_dummies(df.drop(columns="target"))
X_selected = SelectKBest(k=3).fit_transform(X, df["target"])
X_train, X_test, y_train, y_test = train_test_split(X, df["target"])
```

Three leaks

Imputation, category discovery, and supervised selection all happen before the split.



A single pipeline repairs the information flow

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```
from sklearn.feature_selection import SelectKBest
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import make_pipeline

model = make_pipeline(
    prep,
    SelectKBest(k=3),
    LogisticRegression(max_iter=3000)
)
model.fit(raw_train, raw_y_train)
print(model.score(raw_test, raw_y_test))
```

Meaning

Each learned feature step is fitted from the training rows before the test rows are transformed.

Models react differently to the same representation

Representation issue	Linear model	Tree model
feature scale	changes penalty and optimization	usually little effect on split order
monotonic transform	changes linear shape	often preserves order based splits
one hot encoding	creates class specific coefficients	enables categorical branches through indicators
irrelevant columns	can inflate variance	offer extra noisy split candidates

A deployment gate checks more than accuracy

Gate	Question	Evidence
availability	is every feature present at prediction	timestamp and source audit
consistency	does online computation match training	shared code and tests
stability	do distributions and selections persist	time and subgroup monitoring
value	does the feature improve the decision	validated utility and cost

An exam answer should follow the evidence chain

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Open book prompt

A churn model jumps from 0.72 to 0.96 after adding last contact outcome and selecting 30 of 3,000 columns.

- 1 Draw the prediction timeline.
- 2 Audit how selection was fitted.
- 3 Propose a nested evaluation.
- 4 State one operational decision.



Weak answer

Listing definitions without diagnosing the score jump does not solve the case.

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- Strong answer

The diagram illustrates a 'CHECKABLE DECISION LOOP'. At the center is a light blue circle labeled 'DECISION'. Surrounding it are five other circles, each connected to the center and to its neighbors by grey lines, forming a pentagonal loop. The nodes are: 'REPRESENTATION' (top, teal border), 'ASSUMPTIONS' (top-right, purple border), 'CALCULATION' (bottom-right, dark blue border), 'EVIDENCE' (bottom-left, teal border), and 'CONSEQUENCES' (left, red border). The connections form a continuous loop between the outer nodes, with each node also having a direct link to the central 'DECISION' node.

Which representation makes structure learnable without leakage?